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CASE STUDY:1

TOPIC: HOSPITAL READMISSION RECORDS: DATA QUALITY AND WRANGLING

P24CS904

DATA SCIENCE

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CASE STUDY – CS1

HOSPITAL READMISSION RECORDS: DATA QUALITY AND WRANGLING

Domain: Healthcare Analytics
Unit: Unit I
Course Outcomes: CO1, CO6
Sustainable Development Goal: SDG 3 – Good Health and Well-being **Phase:** Phase 1

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1. ABSTRACT

Healthcare organizations generate large volumes of patient information through hospital admissions, laboratory investigations, medication records, and discharge summaries. However, healthcare datasets often contain missing values, inconsistent categories, duplicate records, and formatting errors that affect data analysis and decision-making.

This case study focuses on data quality assessment and data wrangling using the Diabetes 130US Hospitals dataset. The dataset contains 101,766 hospital encounters collected from 130 hospitals in the United States between 1999 and 2008.

The main objective is to identify data quality issues, handle missing values, detect outliers, standardize categorical and numerical attributes, integrate ICD-9 diagnosis information, and transform encounter-level records into a patient-level dataset.

Python, Pandas, NumPy, and Jupyter Notebook are used to perform the data cleaning and transformation processes. The final outcome is a structured, consistent, and analysis-ready healthcare dataset that can support further hospital readmission analysis.

Keywords: Healthcare Analytics, Data Quality, Data Wrangling, Missing Values, ICD-9, Pandas, Hospital Readmission.

2. INTRODUCTION

Healthcare data plays an important role in improving patient care, hospital management, and medical decision-making. Hospitals collect different types of information, including patient demographics, admission details, medical diagnoses, laboratory results, medications, and discharge information.

Raw healthcare data is not always suitable for direct analysis because it may contain missing values, incorrect formats, duplicate entries, and inconsistent data representations.

Data quality and data wrangling are essential processes for converting raw data into reliable and meaningful information.

This case study uses the Diabetes 130-US Hospitals dataset to investigate data quality problems and implement appropriate cleaning techniques. The study also focuses on integrating diagnosis information using ICD-9 codes and restructuring multiple hospital encounters into a patient-level representation.

3. PROBLEM STATEMENT

The raw hospital readmission dataset contains incomplete, inconsistent, and unstructured information that may reduce the accuracy and reliability of healthcare analysis.

Missing values are represented using special symbols such as '?', while some categorical attributes contain inconsistent labels. Numerical attributes may also contain unusual values or outliers.

Additionally, the dataset contains multiple hospital encounters for the same patient, making patient-level analysis difficult.

Therefore, this case study aims to develop a systematic data cleaning and wrangling process to identify and correct data quality issues, standardize the dataset, integrate ICD-9 diagnosis descriptions, and generate a patient-level dataset suitable for further analysis.

4. OBJECTIVES

The major objectives of this case study are:

1. To examine the structure and quality of the hospital readmission dataset.
2. To identify missing values, duplicate records, and inconsistent categories.
3. To detect and investigate unusual numerical values and outliers.
4. To clean and standardize categorical and numerical attributes.
5. To integrate ICD-9 diagnosis codes with their corresponding descriptions.
6. To transform encounter-level records into one row per patient.
7. To document all data cleaning and transformation decisions.
8. To generate a clean dataset suitable for healthcare data analysis.

5. DATASET DESCRIPTION

The dataset selected for this case study is the **Diabetes 130-US Hospitals for Years 1999–2008**, available through the UCI Machine Learning Repository.

Dataset characteristics:

Attribute	Description
Dataset name	Diabetes 130-US Hospitals
Total records	101,766 encounters
Number of hospitals	130
Collection period	1999–2008
Features	47 listed features
Domain	Healthcare
Data format	CSV
Main target	Hospital readmission
Data source	UCI Machine Learning Repository

The dataset includes patient demographics, admission details, length of hospital stay, laboratory procedures, medication information, diagnosis codes, and readmission status.

The readmission variable contains three categories:

- `<30` – Readmitted within 30 days.
- `>30` – Readmitted after 30 days. `NO` – No recorded readmission.

The original dataset contains encounter-level records, meaning that a patient may have multiple hospital visits.

Dataset source:
https://archive.ics.uci.edu/dataset/296/diabetes%2B130us%2Bhospitals%2Bfor%2Byears%2B1999-2008

6. TOOLS AND TECHNOLOGIES

The following tools are used to perform data quality assessment and wrangling.

Tool	Purpose
Python	Main programming language
Pandas	Data manipulation and cleaning
NumPy	Numerical operations
Jupyter Notebook	Interactive coding and analysis
Matplotlib	Data visualization
Seaborn	Statistical visualization

7. DATA QUALITY PROFILING

Data quality profiling is the initial step in understanding the dataset and identifying potential problems.

The following activities are performed:

7.1 Dataset Structure Analysis

- Identify the total number of rows and columns.
- Examine column names and data types.
- Understand the meaning of each attribute.
- Identify unique patient and encounter identifiers.

7.2 Missing Value Identification

The dataset contains missing values represented by the special character `?`.

Important attributes that may contain missing values include:

- Race
- Weight
- Payer code
- Medical specialty
- Diagnosis fields

The values are converted into proper missing-value representations before further processing.

7.3 Duplicate Record Detection

Duplicate records are identified using encounter identifiers and complete-row comparisons.

Duplicate records are investigated before removal to avoid accidentally deleting valid hospital encounters.

7.4 Inconsistent Category Identification

Categorical columns are examined for:

- Extra spaces.
- Different letter cases.

- Unknown or invalid labels.
- Inconsistent category representations.

7.5 Numerical Data Analysis

Numerical attributes such as time in hospital, number of laboratory procedures, number of medications, and previous hospital visits are examined using descriptive statistics.

Minimum, maximum, mean, median, and standard deviation are calculated to understand the distribution of the data.

8. DATA CLEANING AND WRANGLING METHODOLOGY

8.1 Handling Missing Values

Missing values are identified and converted into standard null values.

The following methods are considered:

- Numerical columns: Median imputation where appropriate.
- Categorical columns: Mode imputation or an explicit 'Unknown' category.
- Columns with extremely high missing percentages: Evaluate their usefulness before deciding whether to retain or remove them.
- Missing diagnosis values: Preserve as unknown rather than inventing a medical diagnosis.

The original missing-value information is recorded before applying any cleaning operation.

8.2 Outlier Detection and Treatment

Outliers are identified using the Interquartile Range (IQR) method.

The formula is:

$$IQR = Q3 - Q1$$

$$\text{Lower Bound} = Q1 - 1.5 \times IQR$$

$$\text{Upper Bound} = Q3 + 1.5 \times IQR$$

Values outside these boundaries are investigated.

Since healthcare records may contain genuine extreme observations, outliers are not automatically deleted. Their validity is examined before treatment.

8.3 Numerical Data Normalization

Numerical columns are standardized or normalized when required for further analysis.

Min-Max normalization is calculated using:

$X' = (X - Xmin) / (Xmax - Xmin)$

This converts numerical values into a range between 0 and 1.

8.4 Categorical Data Standardization

Categorical values are standardized by:

- Removing unnecessary spaces.
- Converting labels into a consistent format. □ Handling unknown and invalid categories.
- Maintaining meaningful healthcare classifications.

For example, inconsistent text representations are converted into a single standard category.

8.5 ICD-9 Diagnosis Lookup Integration

The dataset contains three diagnosis columns:

- `diag_1`
- `diag_2`
- `diag_3`

These columns contain ICD-9 diagnosis codes.

An external ICD-9 reference table is integrated with the dataset to obtain the corresponding diagnosis descriptions.

The lookup table contains:

ICD-9 Code	Diagnosis Description
250.00	Diabetes mellitus without mention of complication
401.9	Unspecified essential hypertension
414.01	Coronary atherosclerosis of native coronary artery

The above entries are illustrative examples; the final descriptions must be verified against the selected ICD-9 reference.

The lookup operation uses a left join to preserve the original hospital records even when a diagnosis code has no matching description.

Important: The dataset's `IDs_mapping.csv` describes admission, discharge, and admissionsource identifiers. It is not the ICD-9 diagnosis dictionary. A separate ICD-9 reference table is required.

9. DATA TRANSFORMATION AND RESHAPING

The original dataset contains multiple encounters for the same patient. Therefore, patient-level transformation is performed using the unique patient identifier `patient_nbr`.

The cleaned data is grouped by patient number to create one row per patient.

The following information is considered during aggregation:

- Total number of hospital encounters.
- First and most recent admission information.
- Average hospital stay.
- Total laboratory procedures.
- Total medication count across encounters.
- Previous inpatient visits.
- Most recent diagnosis information. □ Readmission history.

For readmission history, the encounter-level outcomes are retained before aggregation so that the patient-level summary does not lose important information.

The final dataset contains one record per unique patient, with clearly documented aggregation rules.

10. EXPECTED RESULTS AND DISCUSSION

After completing the data cleaning and wrangling process, the following outcomes are expected:

1. Missing values are identified and appropriately handled.
2. Duplicate and inconsistent records are investigated and corrected.
3. Numerical values are examined for outliers.
4. Categorical variables are standardized.
5. ICD-9 diagnosis codes are integrated with readable descriptions.
6. Multiple hospital encounters are summarized into patient-level records.
7. A structured and analysis-ready dataset is generated.

The quality of the dataset is evaluated by comparing the original and cleaned versions.

The following measurements are recorded during implementation:

Evaluation parameter	Before cleaning	After cleaning
Total records	101,766	To be calculated
Total columns	To be verified	To be calculated
Missing values	To be calculated	To be calculated
Duplicate records	To be calculated	To be calculated
Unique patients	To be calculated	To be calculated
Invalid categories	To be calculated	To be calculated

The actual results will be obtained after executing the cleaning procedures using Python.

11. SDG 3 – GOOD HEALTH AND WELL-BEING

This case study supports United Nations Sustainable Development Goal 3, which focuses on ensuring healthy lives and promoting well-being for all.

Reliable healthcare data helps hospitals understand patient records, improve information quality, and support better medical analysis.

By cleaning and organizing hospital readmission records, this project contributes to the development of reliable healthcare information systems and supports future research on patient readmission and hospital care improvement.

12. CONCLUSION

This case study presents a systematic approach to data quality assessment and data wrangling using the Diabetes 130-US Hospitals dataset.

The proposed methodology covers missing value handling, duplicate detection, outlier investigation, numerical normalization, categorical standardization, ICD-9 diagnosis integration, and patient-level data transformation.

Python, Pandas, NumPy, and Jupyter Notebook provide an effective environment for implementing these processes.

The final cleaned dataset will provide a reliable foundation for further healthcare analytics, statistical analysis, and hospital readmission prediction.

13. REFERENCES

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